
ME 631: Heat Transfer and Fluid Flow in Energy Systems

Course Code	ME 631
Course Name	Heat Transfer and Fluid Flow in Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Instructor's consent
Intended for	M. Tech / UG/MS/PhD
Distribution	Core for M. Tech, elective for / UG/MS/PhD
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**

Overview of transport phenomena, Vectors and tensors, types/uses of control volume, notion of conservation principles and constitutive relations [4 Lectures]

- **Module II**

Analysis of flow through pipes, Major and Minor losses, Hydraulic gradient and total energy line – pipe in series and parallel, hydraulic transmission of power, Turbulent flow in pipes, Smooth and rough boundaries, water hammer phenomena, Flow through fluidized and packed bed combustor, flow through nozzle and diffusers, cascade theory, flow through reciprocating, centrifugal and axial pumps, compressors and turbines, Cavitation phenomena. [28 Lectures]

- **Module III**

Heat transfer phenomena in pipes, combustion chambers, fluidized and packed bed combustors, Analysis of heat exchangers, heat and fluid phenomena in boilers. [10 Lectures]

Text and Reference Books

1. William S. Janna, **Design of Fluid and Thermal Systems**, 4th Edition
2. Steven G. Penoncello, **Thermal Energy Systems: Design and Analysis**, CRC Press.
3. R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot, **Transport phenomena**, 2nd Edition, Wiley, 2001.
4. J.C. Slattery, **Advanced transport phenomena**, Cambridge University Press, New York, 1999.
5. Welty J.R, Wicks J.E, Wilson R.E, Rorrer G, **Fundamentals of momentum, heat and mass transport**, 4th Edition, Wiley, 2001.